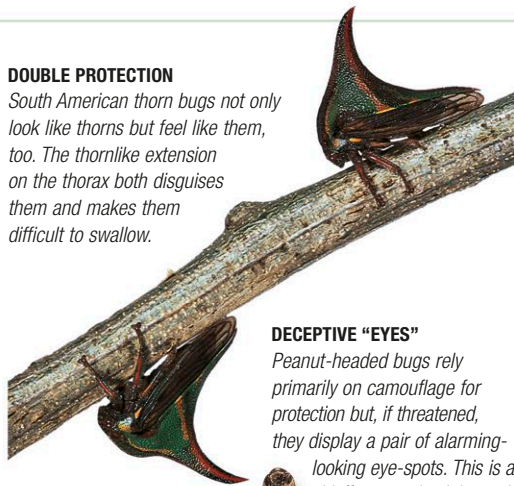


DOUBLE PROTECTION

South American thorn bugs not only look like thorns but feel like them, too. The thornlike extension on the thorax both disguises them and makes them difficult to swallow.

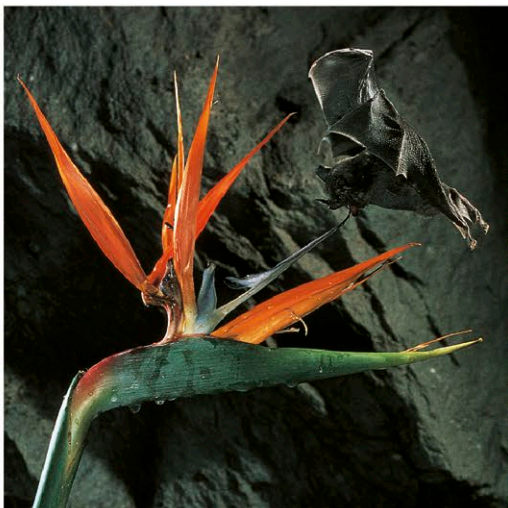
**DECEPTIVE "EYES"**

Peanut-headed bugs rely primarily on camouflage for protection but, if threatened, they display a pair of alarming-looking eye-spots. This is a bluff, suggesting it is much larger than it actually is.

**Feeding**

Near the equator, trees grow, flower, and set seed all year round, generating a nonstop supply of food. Many forest animals—including bats, birds, and insects—live almost exclusively on the abundant nectar and fruit. Some of these animals help trees spread their pollen and seeds. Quetzals, for example, swallow fruit whole, and then regurgitate the pits onto the forest floor where they can germinate.

Compared with flowers and fruit, tropical forest leaves are difficult to digest. Animals that feed on these leaves generally pick them while they are still young—before protective toxins

**NECTAR FEEDERS**

A bat laps up flower nectar. Flowers pollinated by bats and birds have to be robust if they are not to be damaged by their visitors. Birds are attracted to certain flowers mainly by their bright colors, bats by their pungent scent.

RODENT OPPORTUNISTS

Agoutis and other small, ground-dwelling mammals often follow the progress of parrots and monkeys through the trees, gathering the pieces of food that accidentally fall to the ground.



have had a chance to build up inside them. Insects are the most prolific leaf eaters, but some of the forest's larger animals also rely on this difficult diet. They include several kinds of monkeys and sloths, and the hoatzin—a highly unusual bird from South America. The hoatzin processes its food much as a grazing mammal does; after eating, it is often so heavy that it can barely fly.

Tropical forest predators range from some of the world's smallest insects to the largest cats. In an environment that provides lots of cover, most of them stalk their prey rather than running it down. Army and driver ants are the most remarkable exceptions: they hunt in "packs" over 50,000 strong, overpowering and eating anything that cannot escape.

Reproduction

For animals that live in trees, breeding can sometimes involve unusual adaptations. Some tree frogs come down to the forest floor to lay their eggs, but many lay them high up in the canopy, either in water-filled tree holes or in the pools of water that gather in plants. Some frog species are more creative, laying their eggs in nests of foam that keep their eggs moist until the tadpoles are ready to hatch.

Many tropical birds start life in the safety of tree holes, but climbing mammals rarely build nests, and many of their young start life in the open. Young monkeys often cling to their mothers' chests, keeping a tight grip as the parent runs along branches or leaps through the air. For young murine (mouselike) opossums, which live in the American tropics, early life is even more precarious because their mothers do not have well-developed pouches. This means that until their feet have developed the ability to cling, these tiny marsupials hang from their mothers' teats by their mouths, their legs dangling in the air.

**PROTECTIVE NEST**

These African treefrogs have grouped together to make a giant foam nest hanging from a branch. The nest's exterior hardens to protect the eggs and to keep the interior foam moist. When the tadpoles hatch, they break out and drop into water below.

ANT FOLLOWERS

In Central and South American forests, a single column of marauding army ants attracts up to 30 different kinds of birds, all swooping down to catch the tiny animals that burst out of the leaf litter to escape the ants. Some birds flutter ahead of the ant column; others dart among the ants themselves. Some are full-time ant followers, while others follow the throng only for as long as the ants are marching through their territories. A few small animals, including wasps and rove beetles, travel with the army on the ground. Although this sounds dangerous, these animals mimic the ants very closely—an adaptation that prevents them from being attacked while gaining them protection and a supply of free food. Army ants also attract lizards and frogs, as well as parasitic flies that lay their eggs on the fleeing animals.

ARMY ANTS**AERIAL ATTACK**

The blue-crowned motmot, a part-time ant follower, is distantly related to kingfishers. Like them, it swoops down on its prey, carries it back to a perch, and hits it against a branch before eating it.